

AN INTERNSHIP REPORT
ON
STUDENT GRADE MANAGEMENT SYSTEM

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE



By

Roll Number of Student: 2410030408

Name of Student: AAYUSH KUMAR

Batch: 2024 - 2028

CANDIDATE'S DECLARATION

I, AAYUSH KUMAR, hereby declare that the internship report titled "Student Grade Management System" has been completed by me in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Computer Science and Engineering at IILM University, Greater Noida.

The internship project was completed under the CodeZoner Virtual Internship Program, where I worked on the development of a Java-based Student Grade Management System. The work presented in this report is based on the tasks, implementation, testing, documentation, and activities carried out during the internship period.

I further declare that this report is my own work and has not been submitted to any other institute or organization for the award of any degree, diploma, or certificate. All references and resources used during the development of the project have been appropriately acknowledged.

I shall be solely responsible for any kind of copyright violation or misrepresentation of the work presented in this report.

Signature: _____

Student Name: AAYUSH KUMAR

Roll No.: 2410030408

Batch: 2024 - 2028

ACKNOWLEDGEMENT

I am using this opportunity to express my sincere gratitude to everyone who supported me throughout my internship. I am thankful for their guidance, constructive feedback, encouragement, and support during the completion of this work.

I would like to express my gratitude to CodeZoner for providing me with the opportunity to participate in the Virtual Internship Program and gain practical experience in Java programming, data structures, algorithms, software development, testing, documentation, and project management.

I am also thankful to IILM University, Greater Noida, and the faculty members of the School of Computer Science and Engineering for their continuous support and for providing me with the opportunity to undertake this internship as part of my academic curriculum.

I would also like to thank everyone who contributed directly or indirectly to the successful completion of my Student Grade Management System project. Finally, I express my heartfelt gratitude to my parents for their constant support, encouragement, and blessings.

Signature: _____

Student Name: AAYUSH KUMAR

Roll No.: 2410030408

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3. INTERNSHIP COMPLETION CERTIFICATE

Internship Offer Letter



INTERNSHIP OFFER LETTER

Date: 29 July 2026
Ref: CZ/OFL/CZ-2026-0199/2026

Dear Aayush Kumar,

We are delighted to extend this Internship Offer Letter to you for the Data Structures & Algorithms — Java program at CodeZoner. After reviewing your application, we are confident that your skills and enthusiasm make you an excellent fit for this virtual internship opportunity.

INTERNSHIP DETAILS

Intern Name	Aayush Kumar
Student ID	CZ-2026-0199
College / University	IILM UNIVERSITY
Degree Program	B.Tech
Internship Domain	Data Structures & Algorithms — Java
Assigned Project	Student Grade Management System
Internship Mode	Virtual / Work from Home
Duration	4 Weeks (28 Working Days)
Batch	July 2026
Commencement Date	29 July 2026
Completion Deadline	26 August 2026

Achievement Unlocked: +100 XP awarded upon enrollment

Share your offer letter on LinkedIn to earn an additional +75 XP bonus!

Scope of Work & Deliverables

- Design, develop, and document your assigned project independently from start to finish
- Maintain a public GitHub repository with structured commits demonstrating daily progress
- Produce a 3-5 minute professional video demonstration of your completed project
- Submit a comprehensive project report of minimum 100 words explaining your methodology
- Optionally participate in weekly check-ins and CodeZoner community knowledge-sharing sessions

Skills & Competencies You Will Develop

- Real-world project execution under professional timeline constraints
- Version control and collaborative development practices using GitHub
- Technical documentation, presentation, and communication skills
- Problem-solving, debugging, and independent research capabilities

Recognition & Certification

Upon successful completion and submission of the project, the intern will receive a digitally verifiable Certificate of Completion from CodeZoner, valid for LinkedIn and resume.

Please confirm your acceptance by logging into your dashboard and commencing your project.

We are excited to have you as part of the CodeZoner community and wish you great success!

Authorised Signatory

CodeZoner — Virtual Internship Program
contact@codezoner.in | www.codezoner.in
India

Intern's Signature

Date of Acceptance

Certificate of Completion



4. PROJECT DESCRIPTION

4.1 Introduction

This report presents the four-week CodeZoner Virtual Internship undertaken in Java Development, Data Structures and Algorithms. The internship was completed as part of the Bachelor of Technology in Computer Science and Engineering programme at IILM University, Greater Noida.

The main objective was to gain practical Java development experience by building a functional Student Grade Management System. The Java console application manages student information and academic grades. It supports adding, viewing, searching, sorting, updating, calculation of total marks and percentage, grade assignment, transcript generation, input validation, and logging.

The development process moved from environment setup and core implementation to testing, feature improvement, JAR packaging, documentation, and project submission. Git and GitHub were used to keep the project source code and progress documented.

4.2 Organization Profile

CodeZoner is a virtual internship platform that provides structured practical assignments and project-based learning opportunities. For this internship, the platform provided a four-week plan covering Java development, data structures, core implementation, testing, documentation, and final submission. The project-based structure enabled theoretical programming concepts to be applied to a practical software problem.

4.3 Problem Statement

Managing student information and academic grades manually becomes difficult as the number of students increases. Repeated calculation of total marks, percentage, and grades is time-consuming and may lead to human error. The project addressed this need through a simple software system that stores student information and supports basic operations for processing academic records.

4.4 Project Objectives

- Develop a functional Student Grade Management System using Java.
- Apply object-oriented programming through Student, StudentService, and GradeCalculator classes.
- Store and manage records using ArrayList and search students through student IDs.
- Calculate total marks, percentage, and grades automatically.
- Support record sorting, grade updates, validation, and transcript generation.
- Use a menu-driven console interface for simple user interaction.
- Perform unit and performance testing of important operations.
- Use Git and GitHub for version control, documentation, and project management.
- Package and test the application as a Java JAR file.

4.5 Scope of the Project

The project focuses on managing and processing academic information through a Java console application. It is intended as an academic, learning-oriented system; records are stored in memory using ArrayList, so the current version does not provide permanent database storage.

- Adding, viewing, and searching student records.
- Updating student marks and calculating total marks, percentage, and grade.
- Sorting records according to academic performance.
- Generating student transcripts.
- Validating input and handling invalid menu or numerical values.
- Logging selected activities, unit testing, performance testing, and JAR packaging.

Future versions may add database persistence, authentication, a graphical or web interface, and additional academic-management features.

4.6 Technologies and Tools Used

Category	Tool / Technology	Use in Project
Language	Java 17	Application implementation
IDE	IntelliJ IDEA	Coding, debugging, and testing
Data structure	ArrayList	In-memory student records
Version control	Git and GitHub	Source control and documentation
Testing	JUnit 5	Unit testing of key functions

4.7 System Architecture

The application uses a simple layered structure. Main.java displays the menu, accepts input, and calls StudentService. Student.java stores student ID, name, age, course, and subject marks. StudentService.java manages adding, searching, displaying, sorting, updating, and transcript generation using an ArrayList<Student>.

GradeCalculator.java calculates total marks, percentage, and grade. InputValidator.java checks IDs, names, ages, courses, and marks. AppLogger.java records selected activities. JUnit tests cover grade calculation and student-service functionality.

Figure 4.1: Overall system flow

User	Main.java	StudentService	Student / GradeCalculator / InputValidator	Result
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4.8 Methodology

Week 1 - Setup and Research: Installed Java and IntelliJ IDEA, created the GitHub repository, studied data structures and sorting, created the Student class, and implemented basic validation and grade calculation.

Week 2 - Core Implementation: Implemented record storage, sorting, grade updates, invalid-input handling, unit tests, and performance tests. A test using 10,000 records evaluated insertion, sorting, and searching operations.

Week 3 - Advanced Features and Polish: Added transcript generation, improved the console interface, carried out user testing and bug fixing, added logging, and updated documentation.

Week 4 - Packaging and Submission: Packaged and tested the application as a JAR, finalized the README, recorded a demonstration, submitted the project, and performed final testing. The package was executed using `java -jar StudentGradeManagementSystem.jar`.

4.9 Expected Outcomes

- Practical understanding of Java application development and object-oriented design.
- Experience using ArrayList, searching, sorting, validation, and automated grade calculation.
- Experience with JUnit testing, performance testing, debugging, and documentation.
- Experience using Git and GitHub for version control.
- Ability to package and execute a Java application as a JAR file.
- Completion of a functional Student Grade Management System.

4.10 Certificates of Completion and Communication Proof

The Internship Offer Letter and the Internship Completion Certificate are attached in Chapter 3 of this report. The certificate-issue communication proof, confirming that the certificate was officially issued by CodeZoner, is shown below.

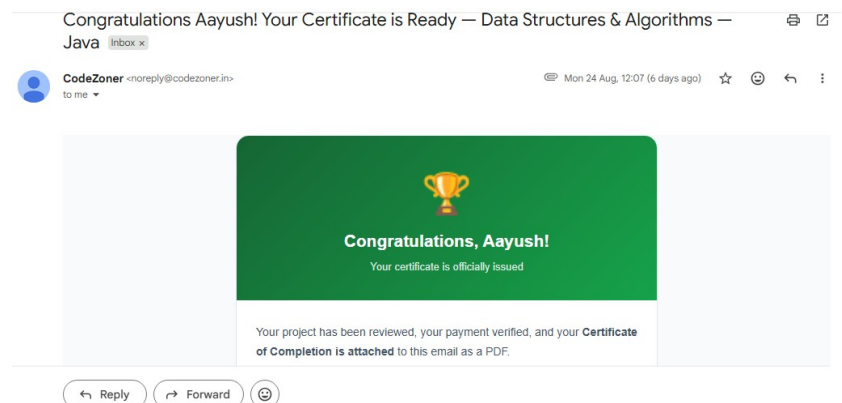


Figure 4.2: CodeZoner email confirming that the certificate has been issued.

5. BIBLIOGRAPHY / REFERENCES

1. CodeZoner, "Java Virtual Internship Program - Student Grade Management System," Internship Tasks and Project Guidelines, 2026.
2. Oracle, "Java Documentation," <https://docs.oracle.com/en/java/>
3. Oracle, "Java Collections Framework," <https://docs.oracle.com/javase/tutorial/collections/>
4. JUnit 5, "JUnit 5 User Guide," <https://junit.org/junit5/docs/current/user-guide/>
5. GitHub Documentation, <https://docs.github.com/>; IntelliJ IDEA Documentation, <https://www.jetbrains.com/help/idea/>